

Project Report

Smart Shop

Management System

Guided By :- Anuj Kumar

Name : Aditya Pal

AFID: AF04971732

Batch Code : ANP-D2405

Course Code : ITPR

Name: Pranjal Khokhar

AFID: AF04991772

Batch Code: ANP-D2405

Course Code : ITPR

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1. Introduction

In the era of digital transformation, retail businesses are rapidly shifting from manual record-keeping to automated systems. Traditional shop management methods suffer from issues such as calculation errors, poor inventory tracking, lack of real-time data, and difficulty in managing taxes like GST. These challenges reduce efficiency and profitability, especially for small and medium-scale retailers.

The **Smart Shop Management System** is a Java-based, console-driven application designed to automate retail operations such as inventory management, customer handling, billing, and sales tracking. The system is developed using **Core Java, JDBC, MySQL, and Maven**, ensuring a professional, scalable, and database-driven architecture. Despite being CLI-based, the application simulates real-world retail workflows accurately and efficiently.

2. Objectives

2.1. Primary Objective

To design and implement a fully functional console-based shop management system that demonstrates real-world software engineering concepts and database integration.

2.2. Specific Objectives

- To automate product, inventory, billing, and customer management.
- To dynamically manage tax configurations (GST slabs).
- To ensure real-time inventory updates after every transaction.
- To implement database normalization and relational integrity.
- To demonstrate practical usage of JDBC, Maven, and SQL.
- To provide a strong academic and industry-oriented project model.

3. Project Category

Application Type: Desktop Console Application (CLI)

Architecture: Layered Architecture (Model, DAO, Utility, Controller)

Database Type: Relational Database Management System (MySQL)

Technology Stack: Java SE, JDBC, MySQL, Maven

4. System Analysis

4.1. Modules and Description

Module 1: Product Management

- Add new products linked with Category, Brand, and Tax
- Update product details (price, brand, tax)
- View product catalog
- Delete obsolete products

Module 2: Inventory Management

- Check available stock
- Refill stock quantities
- Low stock alert system
- Automatic stock deduction after sales

Module 3: Cart & Customer Management

- Customer registration and login
- Product browsing
- Cart operations (add, remove, update quantity)
- Temporary cart handling using Java Collections

Module 4: Billing & Sales Management

- Automatic bill calculation
- Dynamic GST calculation
- Invoice generation
- Sales history storage

4.2. Database Design

The database was implemented using **MySQL 8.0** and consists of **8 normalized tables**. All tables were created programmatically on the first execution of the application using JDBC.

1. **users** – Admin/Staff login credentials

```
mysql> desc users;
```

Field	Type	Null	Key	Default	Extra
user_id	int	NO	PRI	NULL	auto_increment
username	varchar(50)	NO	UNI	NULL	
password	varchar(50)	NO		NULL	
role	varchar(20)	NO		NULL	

2. **categories** – Product categories

```
mysql> desc categories;
```

Field	Type	Null	Key	Default	Extra
category_id	int	NO	PRI	NULL	auto_increment
name	varchar(50)	NO		NULL	
description	varchar(100)	YES		NULL	

3. **brands** – Product brands/manufacturers

```
mysql> desc brands;
```

Field	Type	Null	Key	Default	Extra
brand_id	int	NO	PRI	NULL	auto_increment
brand_name	varchar(50)	NO		NULL	

4. **taxes** – GST and other tax slabs

```
mysql> desc taxes;
```

Field	Type	Null	Key	Default	Extra
tax_id	int	NO	PRI	NULL	auto_increment
tax_name	varchar(20)	YES		NULL	
rate	double	YES		NULL	

5. **products** – Inventory master table

```
mysql> desc products;
```

Field	Type	Null	Key	Default	Extra
product_id	int	NO	PRI	NULL	auto_increment
name	varchar(100)	NO		NULL	
price	double	NO		NULL	
quantity	int	YES		0	
category_id	int	YES	MUL	NULL	
brand_id	int	YES	MUL	NULL	
tax_id	int	YES	MUL	NULL	

6. **customers** – Customer details

```
[mysql> desc customers;
```

Field	Type	Null	Key	Default	Extra
customer_id	int	NO	PRI	NULL	auto_increment
name	varchar(100)	YES		NULL	
phone	varchar(15)	YES	UNI	NULL	
email	varchar(100)	YES		NULL	
password	varchar(50)	NO		NULL	

7. **sales** – Bill header information

```
[mysql> desc sales;
```

Field	Type	Null	Key	Default	Extra
sale_id	int	NO	PRI	NULL	auto_increment
sale_date	datetime	YES		CURRENT_TIMESTAMP	DEFAULT_GENERATED
total_amount	double	YES		NULL	
customer_id	int	YES	MUL	NULL	

8. **cart_items** – Bill details

```
[mysql> desc cart_items;
```

Field	Type	Null	Key	Default	Extra
id	int	NO	PRI	NULL	auto_increment
sale_id	int	YES	MUL	NULL	
product_id	int	YES	MUL	NULL	
qty	int	YES		NULL	
price_at_sale	double	YES		NULL	

Primary and foreign key constraints were enforced to maintain data integrity. Referential relationships were validated through successful CRUD operations during runtime.

4.3 ER Diagram

The ER Diagram represents relationships such as:

Product → Category (N:1) :

- Many Products (e.g., Mouse, Keyboard) belong to One Category (e.g., Electronics).

Product → Brand (N:1)

- Many Products manufactured by One Brand (e.g., Samsung).

Product → Tax (N:1)

- Many Products falls under One Tax Slab (e.g., GST 18%).

Customer → Sales (1:N)

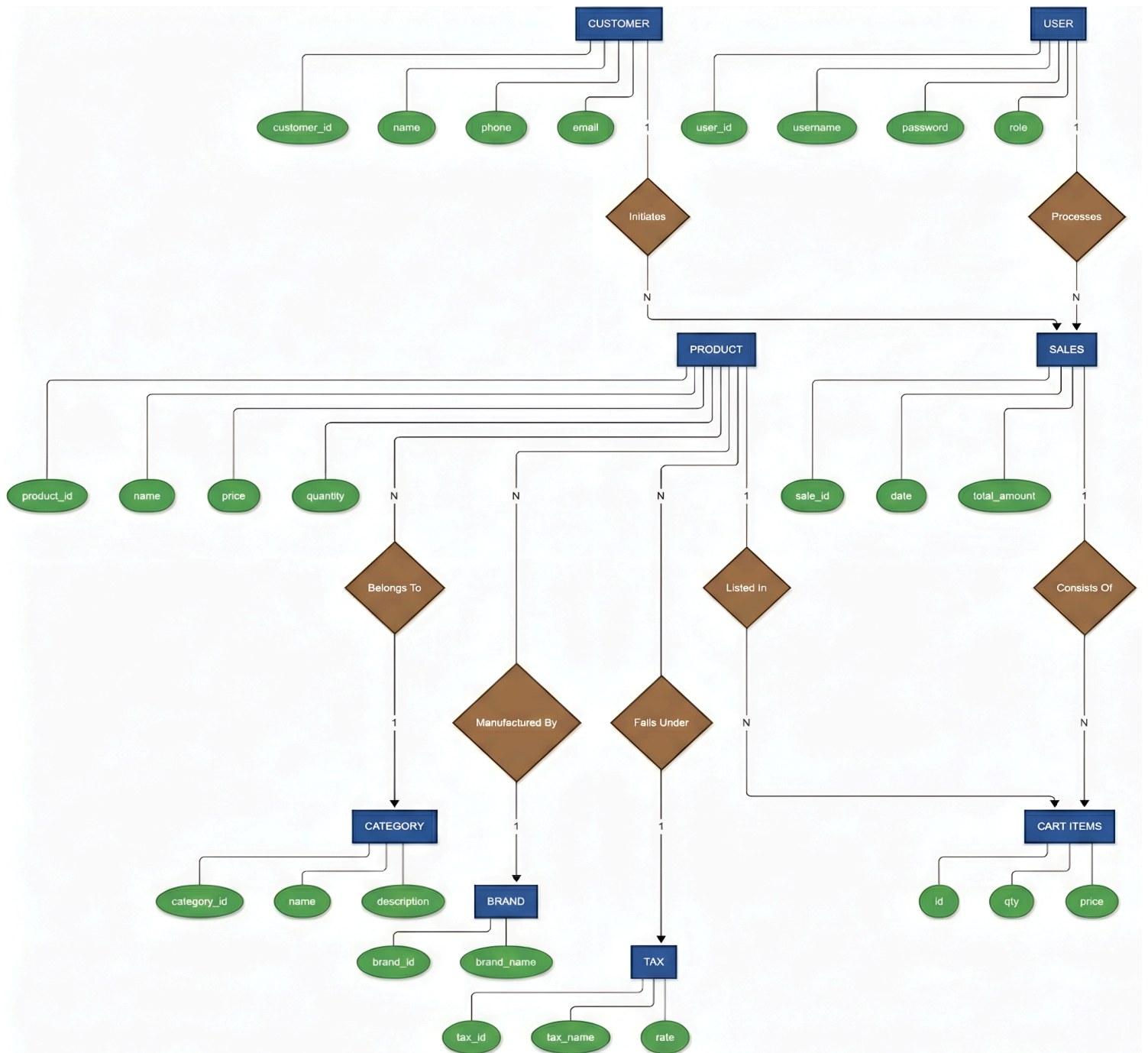
- One Customer *makes* many Sales (Purchases).

Sales → Cart Items (1:N)

- One Sale (Invoice) *contains* many Cart Items (Rows in the bill).

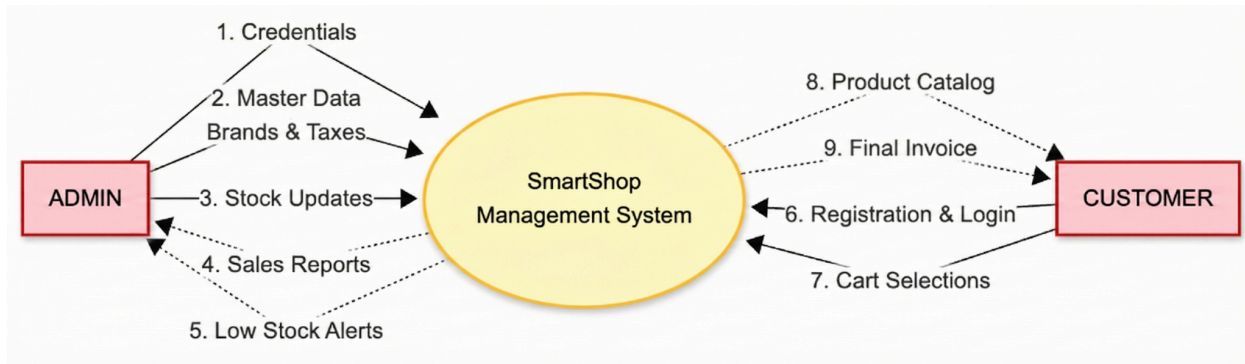
Product → Cart Items (1:N)

- One Product *appears in* many cart Items (Across different bills).

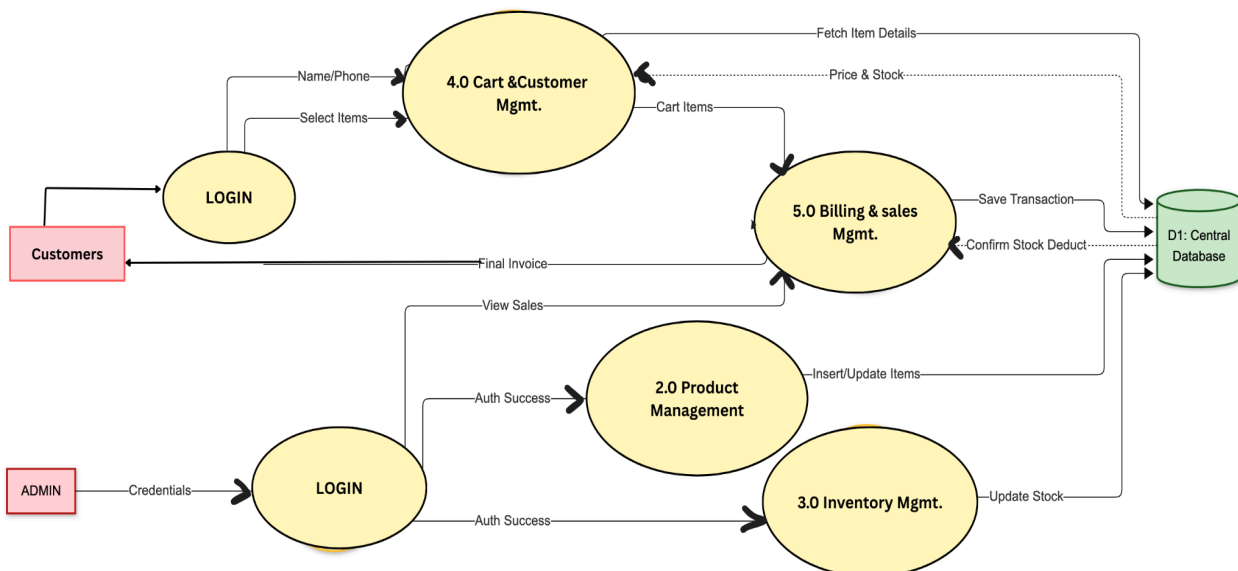


4.4 Data Flow Diagram (DFD)

- **Level 0:** Overall system interaction between Admin, Customer, and Database

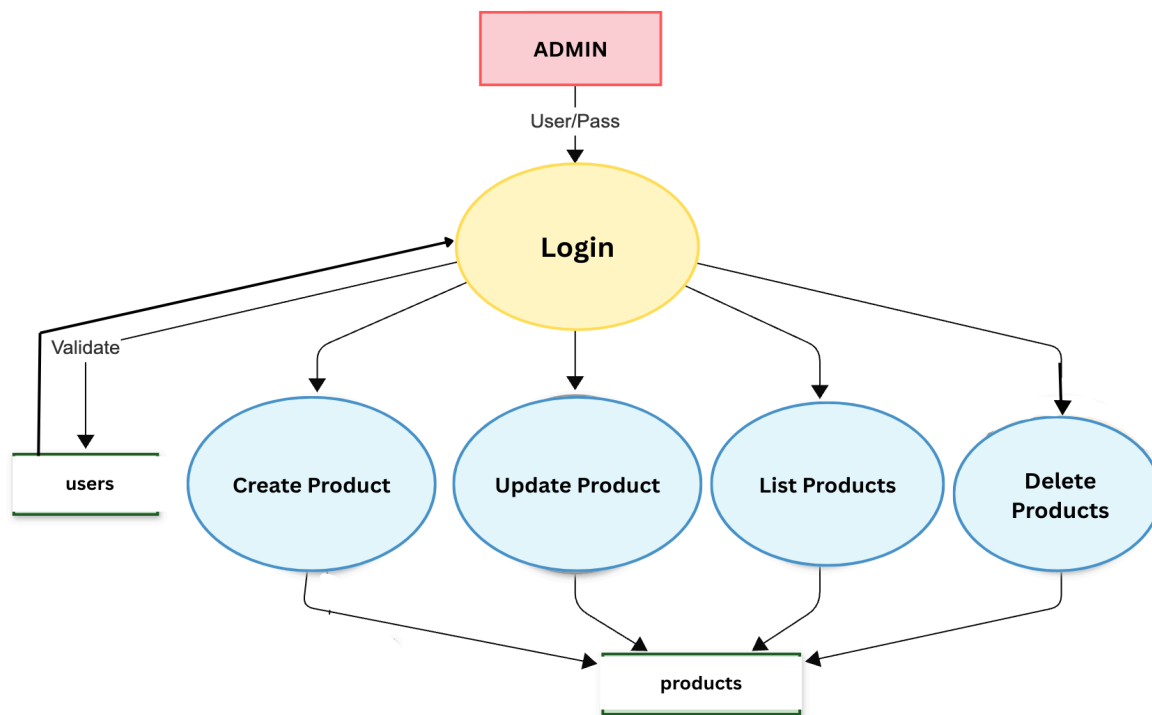


- **Level 1:** Module-wise data flow

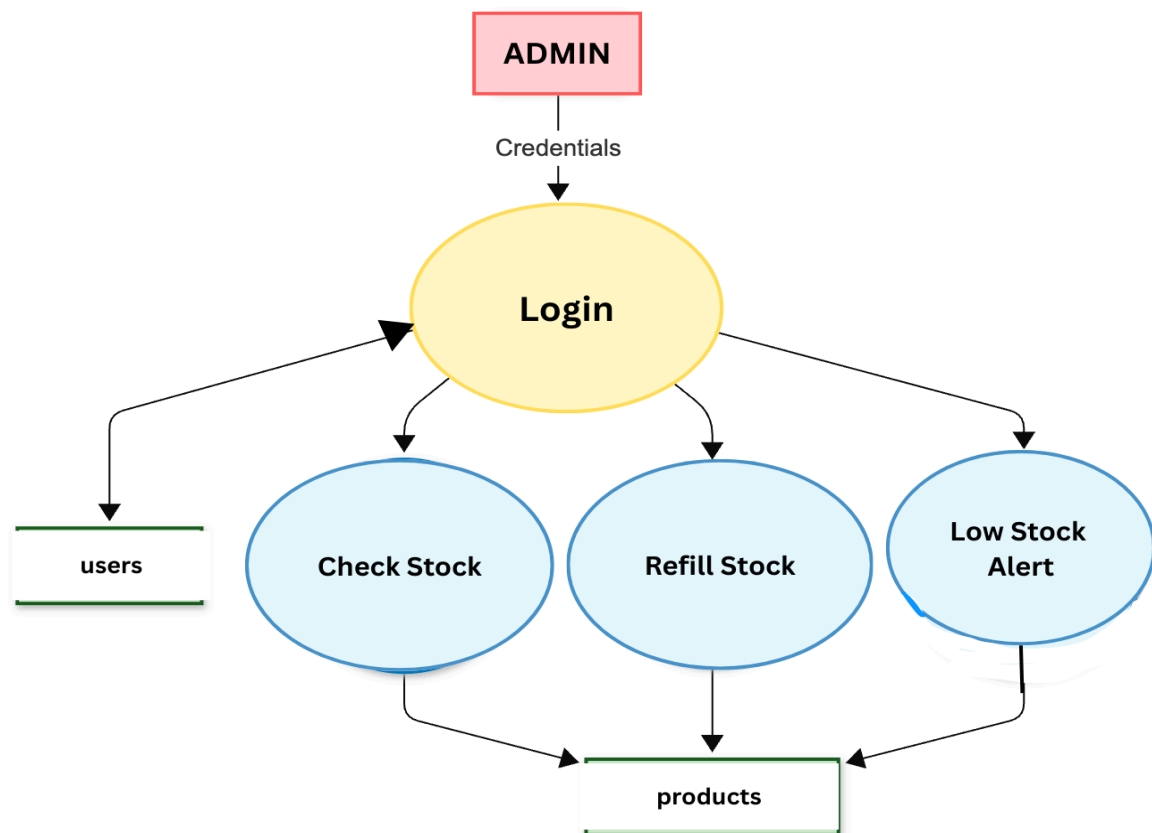


- **Level 2:** Detailed process flow for Product, Inventory, Cart, and Billing modules

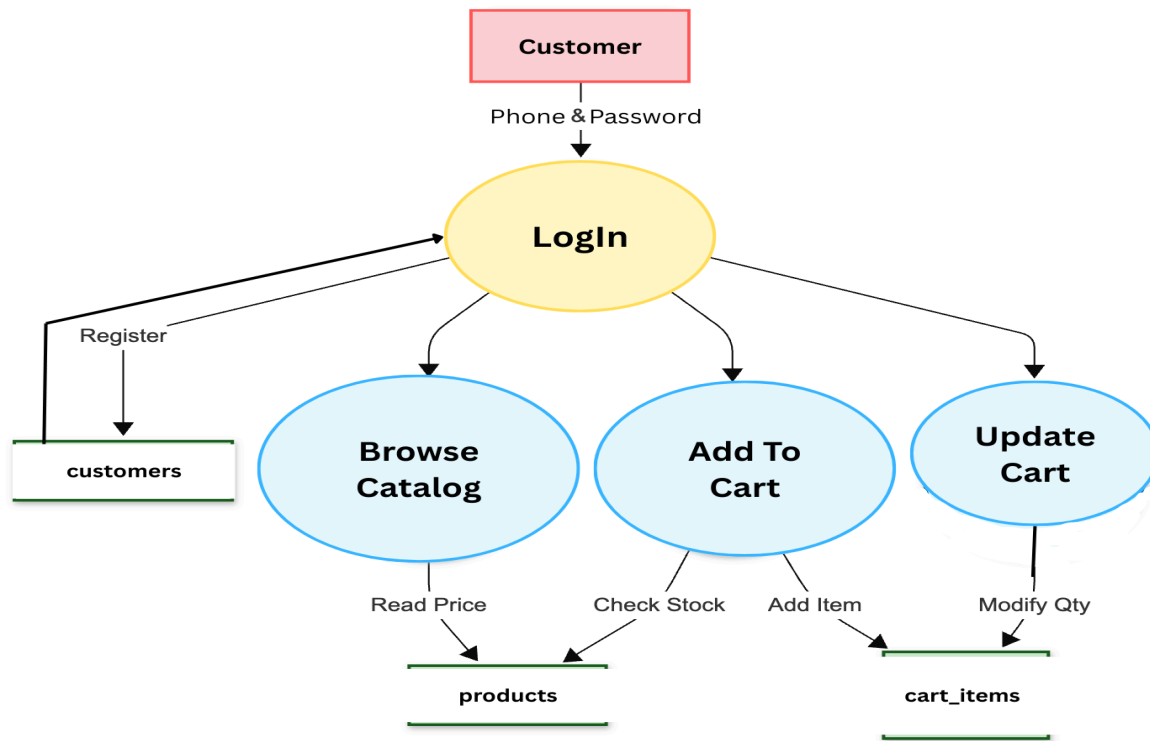
Module 1: Product Management



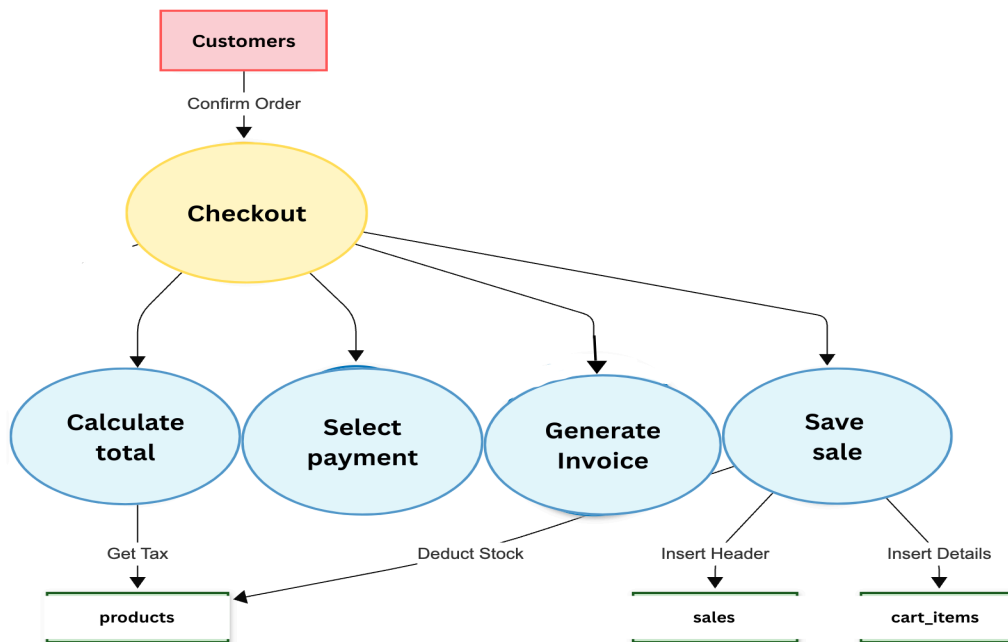
Module 2: Inventory Management



Module 3: Cart & Customer Management



Module 4: Billing & Sales Management





5. Complete System Structure

The project uses a **Layered Architecture** to ensure clean separation of concerns.

5.1 Model Layer

- Represents database tables as Java classes
- Uses encapsulation with getters and setters

5.2 DAO Layer

- Handles all database operations
- Uses JDBC PreparedStatement
- Manages transactions with commit and rollback

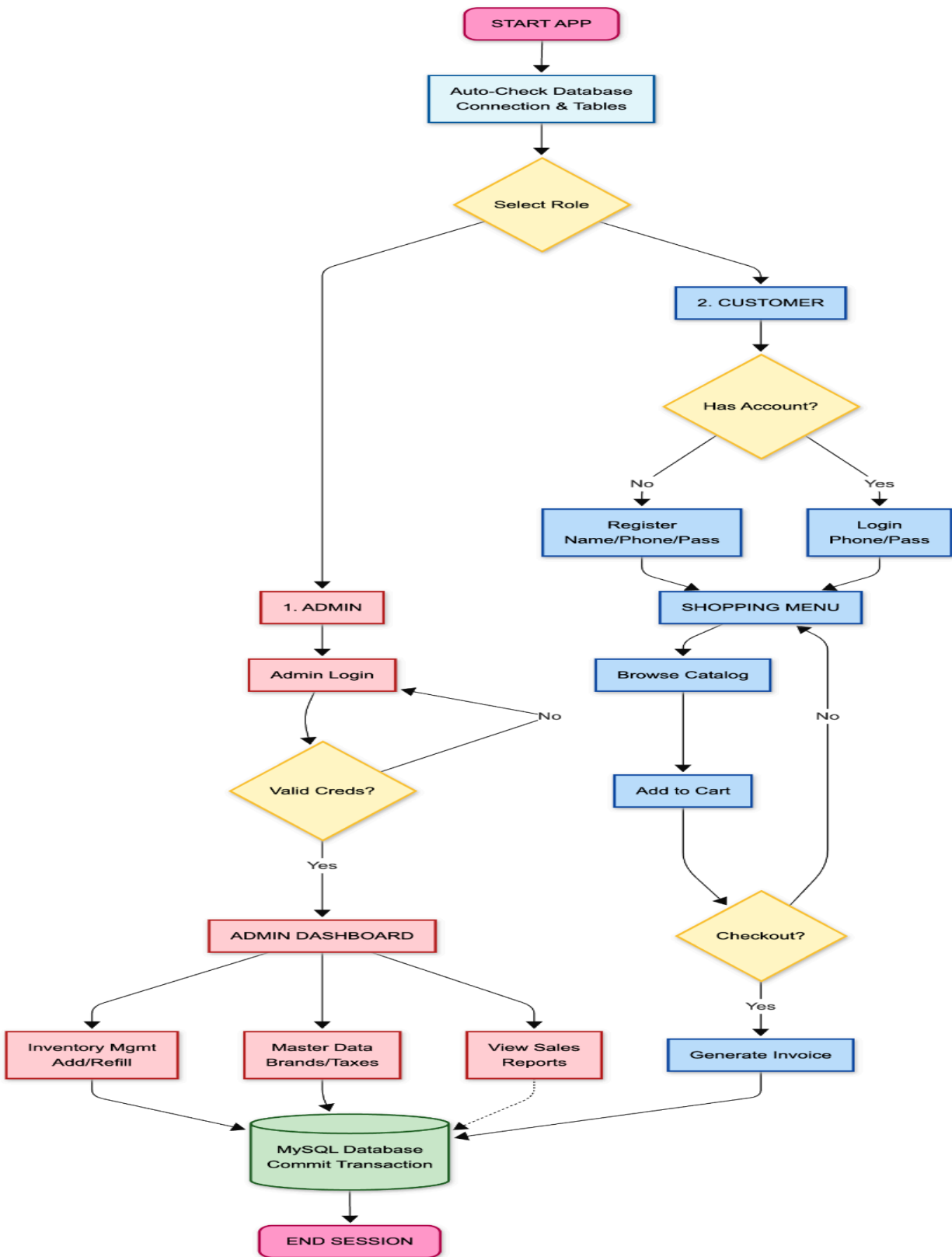
5.3 Utility Layer

- Manages database connectivity
- Automatically creates tables on first run

5.4 Controller Layer

- Application entry point
- Manages user interaction and input validation
- Routes logic between modules

5.1. Process Logical Diagram





6. Testing & Results

- The system was tested using multiple scenarios including:
- Valid and invalid login attempts
- Stock boundary conditions
- Concurrent cart operations
- Successful and failed transactions
- All modules functioned correctly, and the expected outputs were obtained.



7. Challenges Faced

- Managing JDBC transactions across multiple tables
- Maintaining data consistency during billing
- Designing a normalized schema with multiple relationships
- Handling user input validation in a CLI environment

These challenges were resolved through structured design and exception handling.



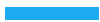
8. Platform Used

8.1 Hardware Requirements

- Processor: Intel Core i3 or higher
- RAM: Minimum 4 GB
- Storage: 500 MB free space

8.2 Software Requirements

- Operating System: Windows / Linux / macOS
- Java SE (JDK 1.4 or above)
- Apache Maven 3.x
- Eclipse IDE
- MySQL Server 8.0
- MySQL JDBC Connector



9. Future Scope

- Migration to Web-based system using Spring Boot
- Barcode scanner integration
- Online payment gateway support
- Graphical dashboards and reports
- More Role-based access control

10. Team Work Distribution

Aditya Pal

- Database Design
- Product Management Module
- Inventory Management Module
- DAO Implementation

Pranjal Khokhar

- Cart & Customer Management Module
- Billing & Sales Module
- Controller Layer

11. Conclusion

The Smart Shop Management System was successfully designed, implemented, and tested. The project demonstrates practical application of Java, JDBC, and database concepts while solving a real-world retail problem. The system is scalable and can be extended to enterprise-level solutions.

12. Bibliography

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- **JDBC MySQL Connector :-**
<https://dev.mysql.com/downloads/connector/j>
- **GeeksforGeeks** – Java & JDBC tutorials
- **MySQL Official Documentation** (Foreign Keys & Normalization).